

# BEAST 3 Migration

Linter & dashboard — state of play

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# Why a migration linter?

- BEAST 3 introduces a **typed spec layer** (`beast.base.spec.*`) replacing legacy `Parameter/Prior/ParametricDistribution` bases.
- ~84 community packages on v2.7 — only ~ 10 have moved to v2.8 so far.
- Per-package migration is mechanical but tedious: hundreds of XML `spec=` attributes, Input declarations, and class bases to update.
- “Encode migration knowledge in the linter, not in chat.” — if a check is mechanical, it should be in a report, not in someone’s head.
- Goal: turn migration into **read the report, apply the hints, re-run**.

# What the tool does

A small Java program (`io.github.beast3.migration`, ~3.2k LOC) that scans sibling checkouts and emits Markdown + JSON.

## Inputs

- `packages.yaml` — 14 tracked packages
- Each package as a sibling checkout
- Maven Central (latest release probe)

## Outputs

- `CHECKLIST.md` — cross-package dashboard
- `reports/<pkg>.md` — per-package, what's left to migrate
- `status.json` — machine-readable
- `BACKLOG.md` — CBAN v2.7 packages not yet on v2.8

Re-run with `mvn -q exec:java`.

# Scanner pipeline

Five phases over each tracked checkout:

1. **Phase 1.** Per package: scan Java classes, XMLs, fxtemplates, Maven build, git commit, Maven Central version.
2. **Phase 2.** Build a **global @Deprecated FQN registry** across all packages + derive spec replacements and unambiguous short-name aliases.
3. **Phase 3.** Chain-resolve each class against the registry — decide `spec` / `mixed` / `legacy` / `neutral`.
4. **Phase 4.** Second XML pass: flag `spec=` and `<map>` bodies pointing at deprecated classes, with replacement hints.
5. **Phase 5.** Java `Input<@Deprecated>` scan — classes whose `Input` type blocks XML migration.

## What the linter checks

- **Class kind tally** per package: Distributions, Operators, Loggers, CalcNodes, Parameters, StateNodes — split into spec / mixed / legacy / neutral.
- **Input rule** (per Kind):
  - Distributions / CalcNodes / StateNodes / Params  $\Rightarrow$  declare *interface* type (RealScalar, RealVector, ...).
  - Operators & Loggers  $\Rightarrow$  may keep concrete RealScalarParam (need `getID()` / write access).
- **Input<@Deprecated>** — a class with an Input on any deprecated type is blocking.
- **XML** — spec / v2.8 / total per file root; deprecated refs in `spec=` or `<map>` bodies.
- **FxTemplates** — clean / spec / total; stay on `version="2.0"` by convention.
- **Build / release** — JPMS module, GH release workflow, CI, Maven Central published version

## Cross-package dashboard (2026-05-13)

| Package           | Stage          | Code | XML | FxT | Dep refs |
|-------------------|----------------|------|-----|-----|----------|
| beast3            | Maven Central  | ●    | ●   | ●   | 637      |
| BEASTLabs         | Maven Central  | ●    | ●   | ●   | 173      |
| beast-classic     | Maven Central  | ●    | ●   | ●   | 112      |
| CoupledMCMC       | Maven Central  | ●    | ●   | ●   | 0        |
| flc               | Maven Central  | ●    | ●   | —   | 0        |
| Mascot            | Maven Central  | ●    | ●   | ●   | 28       |
| morph-models      | Maven Central  | ●    | ●   | ●   | 0        |
| sampled-ancestors | Maven Central  | ●    | ●   | ●   | 0        |
| MutableAlignment  | compile-tested | ●    | ●   | —   | 0        |
| ORC               | Maven Central  | ●    | ●   | ●   | 0        |
| bModelTest        | compile-tested | ●    | ●   | ●   | 0        |
| obama             | compile-tested | ●    | ●   | ●   | 207      |
| feast             | compile-tested | ●    | ●   | ●   | 9        |
| GammaSpikeModel   | compile-tested | ●    | ●   | ●   | 262      |

● legacy / non-migrated   ● minor (concrete Inputs, stray refs)   ● clean   “—” = package does not use beastfx

## State of play — highlights

- 7 of 14 tracked packages are fully green across Code, XML, and FxT: CoupledMCMC, flc, morph-models, sampled-ancestors, MutableAlignment, ORC, bModelTest. flc and MutableAlignment don't use beastfx, so FxT is N/A.
- **beast3 core** is green on Java but still carries 637 deprecated-class references in its example XMLs / fxtemplates — mechanical edits, but many of them.
- **BEASTLabs**, **beast-classic** are the remaining Java-side packages: legacy bases and concrete Input declarations.
- **Mascot** Java side is clean apart from two Loggers carrying `Input<List<Function>>`; 28 deprecated-class references across 14 XMLs and 5 fxtemplates remain — a small task once those Loggers are sorted.
- **obama**, **feast**, **GammaSpikeModel** are the compile-tested-tier packages with work outstanding; obama has 11 legacy CalcNodes and 207 deprecated refs; GammaSpikeModel has 7 classes with Input-rule violations and 262 deprecated refs in XMLs.

# What's blocking what

Visible failure modes the linter surfaces:

- **Concrete-spec Inputs on Distributions/CalcNodes** — BEASTLabs (15/29), obama (9/18), beast-classic (7/15), GammaSpikeModel (7/30). Fix: declare the interface type.
- **Input<@Deprecated>** — a blocking violation, flagged inline in per-package reports.
- **@Deprecated references in XMLs** — the linter flags every spec= attribute and <map> body that names a deprecated class, with the spec replacement hint inline in the per-package report.
- **Loggable classes** — treated as a fallback role, so a class isn't tagged a Logger merely by implementing Loggable.
- **Logger Input rule** — Loggers may declare Input<Function> (they need getID(), which the bare type interfaces don't provide).



## Workflow today

Each migration session looks like:

1. `mvn -q exec:java` — regenerate dashboard + reports.
2. Open `reports/<pkg>.md` — read what's left and the replacement hints.
3. Apply edits in the sibling checkout (Java first, then XMLs, then `fxtemplates`).
4. Re-run linter; iterate until the package goes ● ● ●.
5. Validate XMLs via `mvn exec:exec` on a short-chain run (catches binding bugs the static linter misses).
6. Open PR against the package's repo; check open PRs first (linter only sees the working tree).

## Next 2–4 weeks

- **Complete the dep-ref work in beast3 core** — 637 references to resolve in example XMLs / fxtemplates.
- **BEASTLabs and beast-classic** — complete the Input-rule corrections so downstream packages can build against them.
- **Mascot** — 28 deprecated references across 14 XMLs and 5 fxtemplates; small once the two residual `Input<List<Function>>` Loggers are dealt with.
- **Strengthen the linter**: cross-reference the `@Deprecated` registry against XML `<map>` bodies; add `Input<@Deprecated>` as a column on the dashboard.
- **Triage the 69 untracked v2.7 packages** in BACKLOG.md — decide which to adopt and which to retire.
- **Goal**: every tracked package ● ● ● before the first official BEAST 2.8.0 release (mid-June).

# Summary

- The linter is **the single source of truth** for migration state across 13 packages.
- Seven packages are fully green; what's left in the rest is concrete and mechanically tractable.
- Migration knowledge keeps moving from chat → linter checks → green-light columns.
- Re-run is one Maven command; reports are version-controlled and reviewable.

`github.com/CompEvol/beast3-migration`  
`CHECKLIST.md`    `reports/`    `BACKLOG.md`